

Arduino Project #1

Emsystio

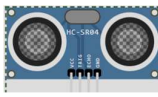
Project Overview:

Controlling the speed of a Servo, with a help of an IR remote control, only operatable within a given range of the distance to the Servo.

Libraries Required:



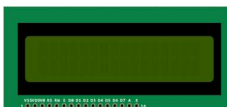
Components Required:



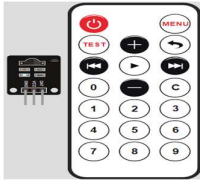
HC-SR04 Ultrasonic sensor



Arduino Mega



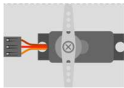
16 x 2 LCD Display



IR Sensor and a Companion remote



Red LED and a 330 ohm resistor



Cross fan Servo



PushDown Button



Potentiometer to connect the LCD

Software Used:

www.Wokwi.com – For code and Connecting Components

Descriptive Working:

When the push button is pressed the lcd display glows on, indicating the user along with the glow of an LED only is he's within the given distance range of –

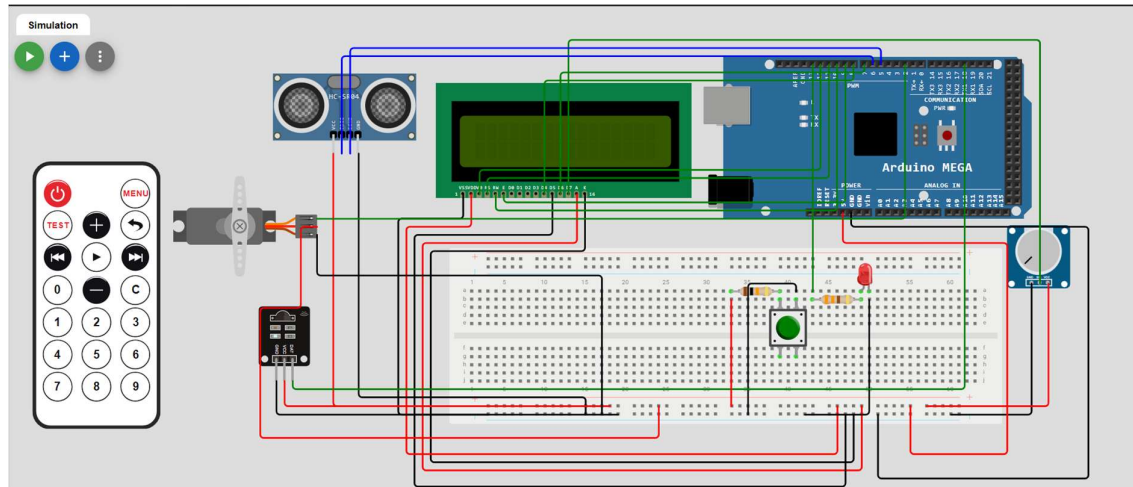
The LED glows indicating that he is well within the range to change the speed of the Servo.

He can operate the servo using the IR remote control.

Practical Use:

Usage in Smart home.

Hardware Connections:



Libraries Version:

```
LiquidCrystal
IRremote@2.6.0
Servo
```

Diagram.json(backend)(by Wokwi):

```
1 {
2   "version": 1,
3   "author": "Anonymous maker",
4   "editor": "wokwi",
5   "parts": [
6     {
7       "type": "wokwi-breadboard",
8       "id": "bb1",
9       "top": 102.6,
10      "left": 233.2,
11      "attrs": { "color": "#eeefed" }
12    },
13    { "type": "wokwi-arduino-mega", "id": "mega", "top": -124.2, "left": 534, "attrs": {} },
14    { "type": "wokwi-hc-sr04", "id": "ultrasonic1", "top": -104.1, "left": 24.7, "attrs": {} },
15    { "type": "wokwi-lcd1602", "id": "lcd1", "top": -92.57, "left": 217.6, "attrs": {} },
16    { "type": "wokwi-ir-receiver", "id": "ir1", "top": 171.45, "left": 7.82, "attrs": {} },
17    { "type": "wokwi-ir-remote", "id": "remote1", "top": 153.6, "left": -163.2, "attrs": {} },
18    {
19      "type": "wokwi-led",
20      "id": "led1",
21      "top": 111.6,
22      "left": 705,
23      "attrs": { "color": "red", "flip": "1" }
24    },
25    {
26      "type": "wokwi-resistor",
27      "id": "r1",
28      "top": 157.55,
29      "left": 662.4,
30      "attrs": { "value": "330" }
31    },
```

```
32    {
33      "type": "wokwi-servo",
34      "id": "servo1",
35      "top": 16.4,
36      "left": -93.2,
37      "rotate": 180,
38      "attrs": { "horn": "double" }
39    },
40    {
41      "type": "wokwi-pushbutton",
42      "id": "btn1",
43      "top": 172.2,
44      "left": 598.2,
45      "rotate": 90,
46      "attrs": { "color": "green", "xray": "", "bounce": "1", "key": "1" }
47    },
48    { "type": "wokwi-potentiometer", "id": "pot1", "top": 75.5, "left": 892.6, "attrs": {} },
49    {
50      "type": "wokwi-resistor",
51      "id": "r2",
52      "top": 147.95,
53      "left": 566.4,
54      "attrs": { "value": "10000" }
55    },
56  ],
57  "connections": [
58    [ "pot1:GND", "bb1:bn.49", "black", [ "v0" ] ],
59    [ "pot1:VCC", "bb1:bp.46", "red", [ "v0" ] ],
60    [ "mega:5V", "bb1:bp.45", "red", [ "v35.7", "h194.3", "v249.6", "h-105.6" ] ],
61    [ "mega:GND.2", "bb1:bn.41", "black", [ "v16.5", "h280.55", "v297.6", "h-240" ] ],
62    [ "pot1:SIG", "lcd1:D7", "green", [ "v-288", "h-557.2" ] ],
```

```

63 [ "lcd1:K", "bb1:bn.39", "black", [ "v67.2", "h-182.4", "v249.6", "h489.6" ] ],
64 [ "lcd1:A", "bb1:bp.40", "red", [ "v57.6", "h-182.5", "v268.8", "h508.8" ] ],
65 [ "lcd1:D6", "mega:7", "green", [ "v-144", "h345.8" ] ],
66 [ "lcd1:D5", "bb1:bn.38", "black", [ "v48", "h-163.1", "v297.6", "h499.2" ] ],
67 [ "lcd1:D4", "mega:8", "green", [ "v-134.4", "h355.2" ] ],
68 [ "lcd1:VDD", "bb1:bp.37", "red", [ "v38.4", "h-76.7", "v268.8", "h499.2" ] ],
69 [ "lcd1:VSS", "bb1:bn.1", "black", [ "v28.8", "h-76.8", "v268.8" ] ],
70 [ "lcd1:E", "mega:9", "green", [ "v9.6", "h384" ] ],
71 [ "lcd1:RW", "mega:10", "green", [ "v19.2", "h393.5" ] ],
72 [ "lcd1:RS", "mega:11", "green", [ "v-19.2", "h393.9" ] ],
73 [ "lcd1:V0", "mega:12", "green", [ "v-28.8", "h393.3" ] ],
74 [ "mega:13", "bb1:43t.a", "green", [ "v0" ] ],
75 [ "r1:1", "bb1:43t.b", "", [ "$bb" ] ],
76 [ "r1:2", "bb1:49t.b", "", [ "$bb" ] ],
77 [ "led1:A", "bb1:49t.a", "", [ "$bb" ] ],
78 [ "led1:C", "bb1:50t.a", "", [ "$bb" ] ],
79 [ "bb1:50t.b", "bb1:bn.34", "black", [ "v0" ] ],
80 [ "bb1:41t.a", "bb1:bn.28", "black", [ "v-9.6", "h9.6" ] ],
81 [ "ultrasonic1:GND", "bb1:bn.11", "black", [ "v288", "h123.6" ] ],
82 [ "ultrasonic1:VCC", "bb1:bp.3", "red", [ "v0" ] ],
83 [ "ultrasonic1:ECHO", "mega:5", "blue", [ "v-124.8", "h632.8" ] ],
84 [ "ultrasonic1:TRIG", "mega:6", "blue", [ "v-115.2", "h623.6" ] ],
85 [ "ir1:GND", "bb1:bn.15", "black", [ "v0" ] ],
86 [ "ir1:VCC", "bb1:bp.14", "red", [ "v0" ] ],
87 [ "ir1:DAT", "mega:18", "green", [ "v9.6", "h787.2" ] ],
88 [ "servo1:GND", "bb1:bn.13", "black", [ "h0", "v48", "h326.4" ] ],
89 [ "servo1:V+", "bb1:bp.19", "red", [ "h-19.2", "v95.9", "h-48", "v153.6", "h470.4" ] ],
90 [ "servo1:PWM", "mega:2", "green", [ "h0" ] ],
91 [ "r2:1", "bb1:33t.a", "", [ "$bb" ] ],
92 [ "r2:2", "bb1:39t.a", "", [ "$bb" ] ],
93 [ "bb1:33t.b", "bb1:bp.26", "red", [ "v0" ] ],

```

```

92 [ "r2:2", "bb1:39t.a", "", [ "$bb" ] ],
93 [ "bb1:33t.b", "bb1:bp.26", "red", [ "v0" ] ],
94 [ "bb1:39t.d", "mega:3", "green", [ "v-249.6", "h134.4" ] ],
95 [ "btn1:1.l", "bb1:41t.b", "", [ "$bb" ] ],
96 [ "btn1:2.l", "bb1:39t.b", "", [ "$bb" ] ],
97 [ "btn1:1.r", "bb1:41b.g", "", [ "$bb" ] ],
98 [ "btn1:2.r", "bb1:39b.g", "", [ "$bb" ] ]
99 ],
100 "dependencies": {}
101 }

```

Frontend Code:

```
//Those seeing from Github: Advised to see from Laptop
// Project description: Project on Remote controlling a servo
//C ++ Code
//Intsalling Libraries
#include <LiquidCrystal.h>
#include <Servo.h>
#include <IRremote.h>

//Creating Variables
int j;
int speed;
String input;
String msg="Enter your name: ";
int IRpin=18;
String Remotecontrol;
decode_results cmd;
IRrecv IR(IRpin);
Servo myServo;
LiquidCrystal lcd(7,8,9,10,11,12);
int swival;
int ultratime;

void setup(){
  lcd.begin(16,2);
  myServo.attach(2);
  IR.enableIRIn();
  Serial.begin(9600);
  pinMode(3,INPUT);
  pinMode(13,OUTPUT);
  pinMode(5,INPUT);
  pinMode(6,OUTPUT);
}
```

```
34 void loop(){
35     while(IR.decode(&cmd)==0){
36
37     }
38     IR.resume();
39     Serial.println(cmd.value,HEX);
40
41
42     Serial.println(msg);
43     while(Serial.available()==0){
44
45     }
46
47     digitalWrite(6,LOW);
48     delayMicroseconds(10);
49     digitalWrite(6,HIGH);
50     delayMicroseconds(10);
51     digitalWrite(6,LOW);
52     ultratime=pulseIn(5,HIGH);
53     delay(25);
54     input=Serial.parseInt();
55     swival=digitalRead(3);
56     if(swival==0){
57         lcd.setCursor(0,0);
58         lcd.print("Hello",input);
59         delay(1000);
60         lcd.clear();
61         if(ultratime < 300){
62             lcd.setCursor(0,0);
63             lcd.print("Set speed Remote");
```

```

65 //When the user clicks 1 in the remote
66 if (cmd.value==0xFF0CF3){
67     for (j=1; j<= 18; j=j+1){
68         myServo.write(20);
69         delay(100);
70     }
71 }
72
73 //When the user clicks 2 in the remote
74 if (cmd.value== 0xFF18E7){
75     for (j=1; j<=12; j=j+1){
76         myServo.write(30);
77         delay(100);
78     }
79 }
80
81 if(cmd.value==0xFF5EA1){
82     for(j=1; j<=6; j=j+1){
83         myServo.write(60);
84         delay(100);
85     }
86 }
87
88 if(cmd.value==0xFF08F7){
89     for(j=1; j<=3; j=j+1){
90         myServo.write(120);
91         delay(100);
92     }

```

```

95     if(cmd.value==0xFF1CE3){
96         for(j=1; j<=2; j=j+1){
97             myServo.write(180);
98             delay(100);
99         }
100     }
101 }
102 }
103 }
104 }
105 }
106 }
107 }
108 }
109 }

```